Ameen Ahmed

01/25/2026

Twilight Wolfpack Analytics – Point Spread Prediction

Overview

For this project, we developed a baseline statistical model to predict point spreads for 78 upcoming ACC men's basketball games. The point spread is defined as the difference between the home team's score and the away team's score. Our approach uses historical game outcomes to estimate relative team strength and home-court advantage in a transparent and interpretable way.

Data

We used NCAA men's basketball game data from the 2021-2022, 2022–2023, 2023–2024, and the 2024–2025 seasons. Only completed games with non-missing scores were included.

Training data initially focused on ACC conference games to understand point spread behavior within a consistent competitive environment. For the final model used in prediction, we expanded the dataset to include all completed games involving ACC teams, including non-conference matchups. This ensured that strength estimates were available for both ACC teams and any non-conference opponents appearing in the competition schedule.

For each game, the point spread was calculated as home points minus away points.

Model

We fit a linear regression model of the form:

$$pt_{\text{spread}} = \beta_0 + \beta_{\text{home team}} + \beta_{\text{away team}} + \epsilon$$

Home and away teams were treated as categorical variables. This structure allows the model to capture baseline scoring tendencies as well as team-specific home and away effects. Model coefficients are interpretable as average point contributions relative to the baseline.

Prediction Process

The fitted model was applied to the official competition template. Because team names in the template used shortened forms, a controlled name-mapping procedure was implemented to align them with the full team names used during model training. Predictions were generated for all 78 games, rounded to two decimal places, and all required metadata fields were completed according to submission guidelines.